

ME101: Engineering Mechanics, 2023-24

IIT Guwahati

Tutorial – 12 (Question) on 22.04.2024

Q1: The uniform slender bar of mass ρ per unit length is freely pivoted about the y axis at the clevis (Fig.1), which rotates about the fixed vertical z -axis with a constant angular velocity ω . Determine the steady-state angle θ assumed by the bar. Length b is greater than length c . [15 Marks]

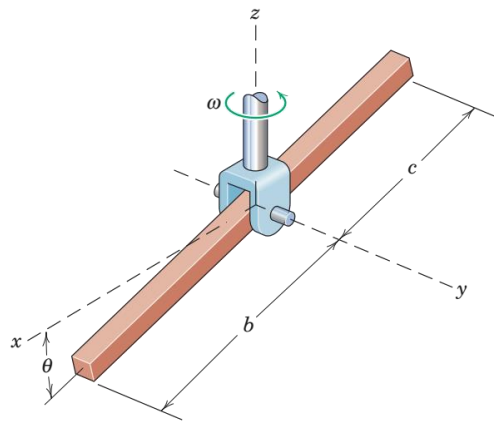


Figure 1

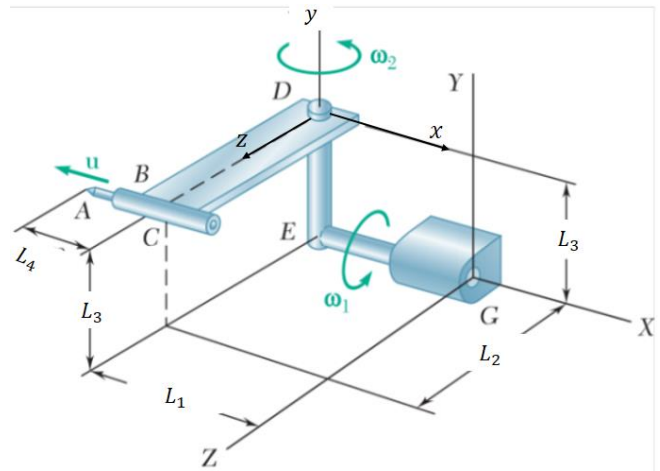


Figure 2

Q2. The position of the stylus tip A is controlled by the robot shown in Fig. 2. In the position shown the stylus moves at a rate of speed $u = 0.2 \text{ m/s}$ which reduces at a rate of 0.3 m/s^2 relative to the solenoid BC . At the same time, arm CD rotates at the constant rate $\omega_2 = 3 \text{ rad/s}$ with respect to component DEG . Knowing that the entire robot rotates about the X -axis at rate $\omega_1 = 5 \text{ rad/s}$ increases at the rate of $\alpha_1 = 2 \text{ rad/s}^2$ determine (a) the velocity of A , (b) the acceleration of A . Take $L_1 = 750 \text{ mm}$, $L_2 = 1000 \text{ mm}$, $L_3 = 500 \text{ mm}$, $L_4 = 300 \text{ mm}$. [15 Marks]

Q3. The centre O of the spacecraft is moving through space with a constant velocity of $8\hat{k} \text{ m/s}$ as shown in Fig. 3. During the period of motion prior to stabilization, the spacecraft has a constant rotational rate $\Omega = 5 \text{ rad/sec}$ about its z -axis. The x - y - z axes are attached to the body of the craft, and the solar panels rotate about the y -axis at the constant rate $\dot{\theta} = 2 \text{ rad/sec}$ with respect to the spacecraft. If ω is the absolute angular velocity of the solar panels, determine ω . Also find the velocity and acceleration of point A when $\theta = 30^\circ$. Take $L = 1 \text{ m}$, $t = 0.2 \text{ m}$. [10 Marks]

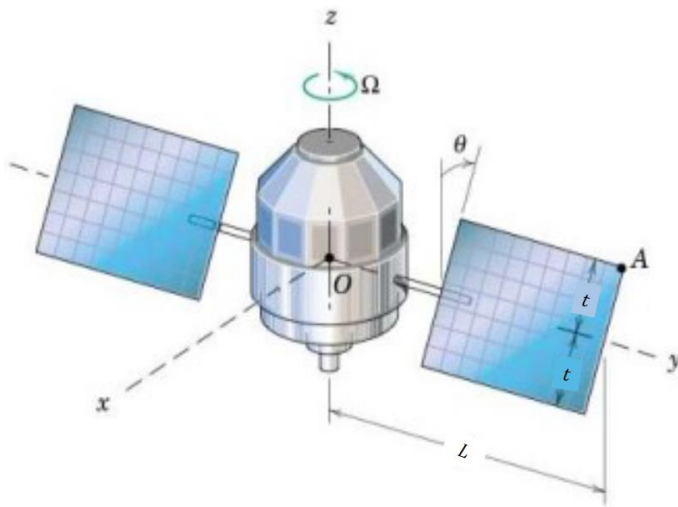


Figure 3

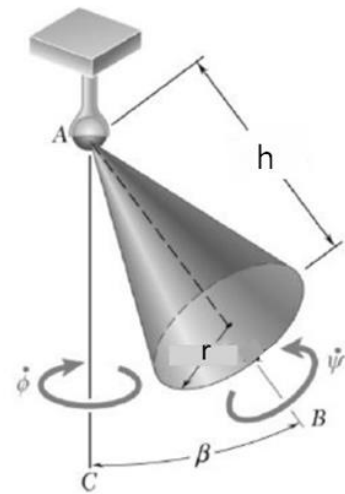


Figure 4

Q4. A solid cone of height h (300 mm) with a circular base of radius r (100 mm) is supported by a ball-and-socket joint at A as shown in Fig. 4. Knowing that the cone is observed to precess about the vertical axis AC at the constant rate of 40 rpm in the sense indicated and that its axis of symmetry AB forms an angle $\beta = 40^\circ$ with AC, determine the rate at which the cone spins about the axis AB. [10 Marks]

Q5. The plate, shown in Figure 5, has a mass of 3 kg and is welded to the fixed vertical shaft, which rotates at the constant speed of 20π rad/s. Compute the moment M applied to the shaft by the plate due to dynamic imbalance. [10 Marks]

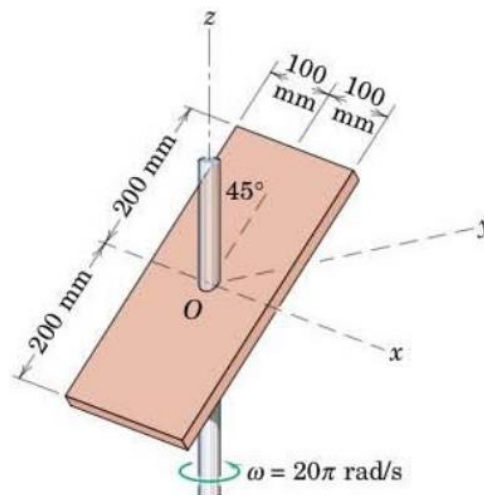


Figure 5